Madhuri Hegde

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F-1 Student | CPT Eligible Summer 2027

EDUCATION

Northeastern University, Silicon Valley — San Jose, CA
M.S. in Computer Science — *Sep 2026 - May 2028 (expected)*
Coursework: Programming Design Paradigm, Database Design

PES University — Bangalore, India
Master of Computer Applications | 8.10/10.0 — *Jan 2021 - May 2022*
Coursework: Data Structures, Design & Analysis of Algorithms, Object-Oriented Modeling & Design, Software Engineering, Cloud Computing

JSS SMI UG and PG Studies — Dharwad, India
Bachelor of Computer Applications — *Jun 2017 - Sep 2020*

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL, Shell Scripting

Software Engineering: Distributed Systems, Microservices, REST APIs, JUnit, pytest, Unix/Linux, Agile

Cloud & DevOps: AWS (EKS, EC2, S3), GCP (GKE, GCS), Docker, Kubernetes, Terraform, Git/GitHub, CI/CD

Web: React, Node.js, Express, WebSockets, HTML/CSS

Data & Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis, Teradata, Kafka, ETL/ELT, Airflow, Data Modeling, Great Expectations

AI-Assisted Development: Claude Code, LLM-assisted development

PROFESSIONAL EXPERIENCE

Rakuten India — Bangalore, India
Data Engineer — *Jun 2022 - Mar 2025*

- Designed and shipped a containerized data preprocessing service on AWS EKS, owning 2 of 5 processing modules end to end, including the REST API layer, transformation logic, and Terraform provisioned infrastructure for source specific ingestion of structured and unstructured data from S3.
- Extended the service by turning recurring ad hoc requests from data science partners into reusable features, removing 6 to 10 hours of manual rework per week.
- Hardened the same service in production, halving recurring downstream incidents by automating data validation and debugging failures at the source.
- Cut daily processing runtime ~40%, from a full workday to roughly five hours, by eliminating full table scans in Teradata joins, partitioning hot tables, and parallelizing previously serial pipeline stages.
- Managed 15 production data pipelines processing 5M+ records daily across AWS/GCP using Python, SQL, and Airflow; debugged production issues and resolved failures across e-commerce, sports, and web traffic workloads.
- Mentored 3 junior engineers through code reviews focused on API and database design, test coverage, and implementation quality.

Rakuten India — Bangalore, India
Tech Intern — *Jan 2022 - Jun 2022*

- Built an automated data validation framework in Python with Great Expectations and auto-generated Data Docs, giving the team direct visibility into pipeline correctness; converted to full-time Data Engineer.

PROJECTS

AI Interview Platform | React, Node.js, Express, PostgreSQL, pgvector, Kafka, Docker, AWS EC2, Claude Code

- Generates role specific interview questions from a user resume and job description, then runs an interactive mock interview with real-time answer evaluation and tailored follow-ups.
- Built an event-driven RAG interview engine on Kafka, retrieving embedded resume and job description chunks from pgvector to ground follow-up questions, with answer scoring consumed asynchronously so evaluation never blocks the interview flow.

Foodie | React, Node.js, Express, MongoDB, PostgreSQL, Redis, WebSockets

- Designed a horizontally scalable real-time chat service using WebSockets, Redis Pub/Sub, and PostgreSQL, supporting message persistence and cross server message delivery.
- Built recipe search and curation features on a React frontend backed by Express and MongoDB.

ACHIEVEMENTS

Incredible Team Award (MVP), Rakuten | MEP Award, Rakuten | MRD Scholarship, PES University